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(54) **WIRE JIG AND TABBING APPARATUS INCLUDING THE SAME**

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B23K 3/08 (2006.01)
H01L 23/00 (2006.01)
B23K 101/38 (2006.01)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Provided are a wire jig and a tabbing apparatus including the wire jig. The wire jig configured to support at least one wire includes: a main body including a window therein; and at least one support portion arranged to correspond to the window and including at least one support component, wherein the at least one support component is configured to press the at least one wire.

13 Claims, 8 Drawing Sheets

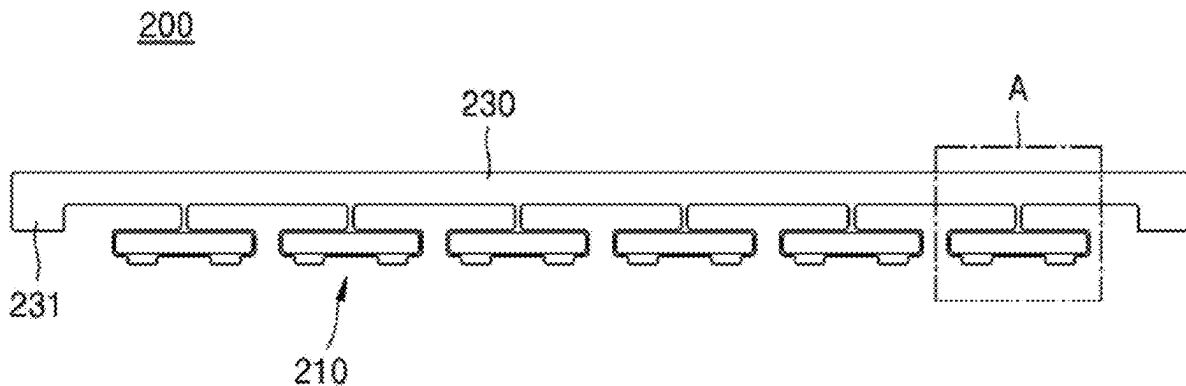


FIG. 1

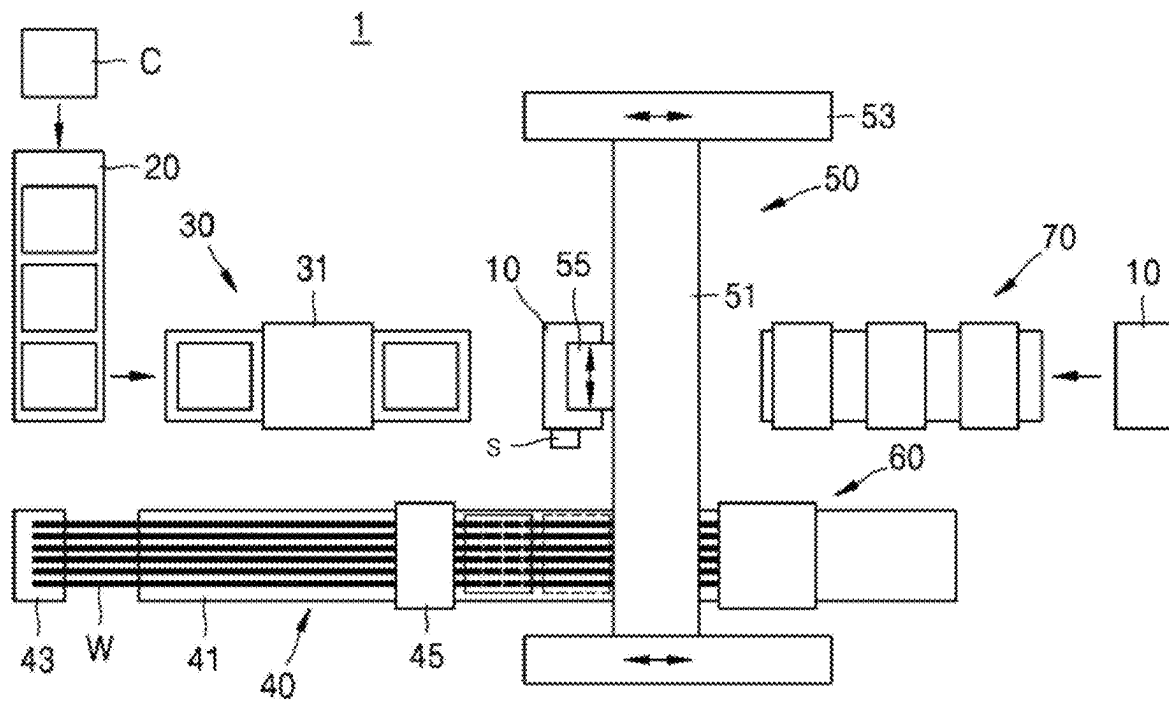


FIG. 2

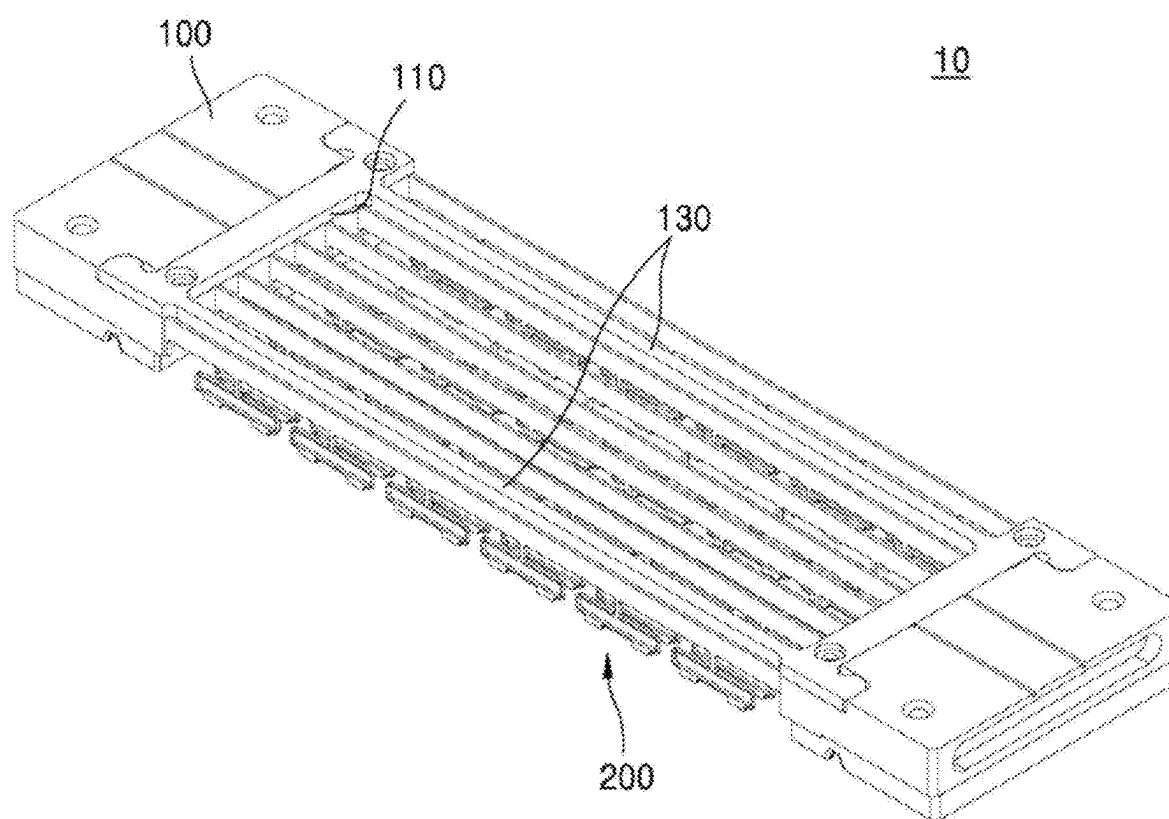


FIG. 3

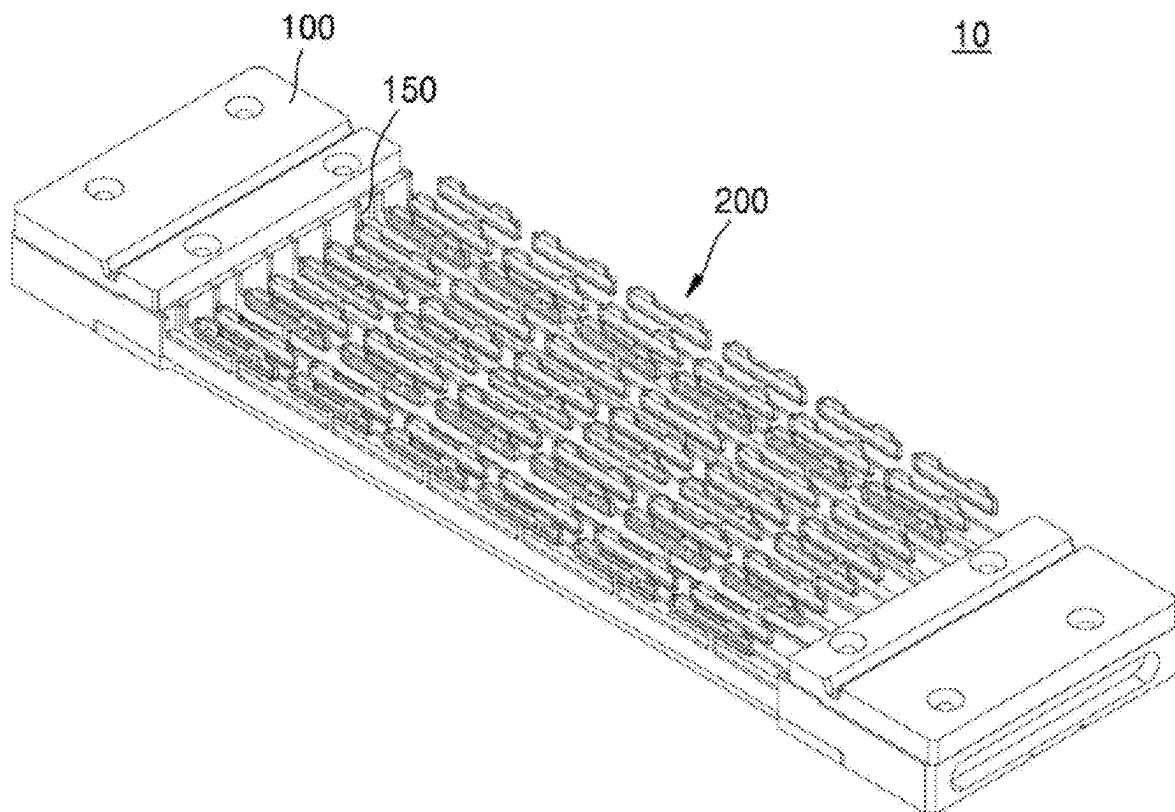


FIG. 4

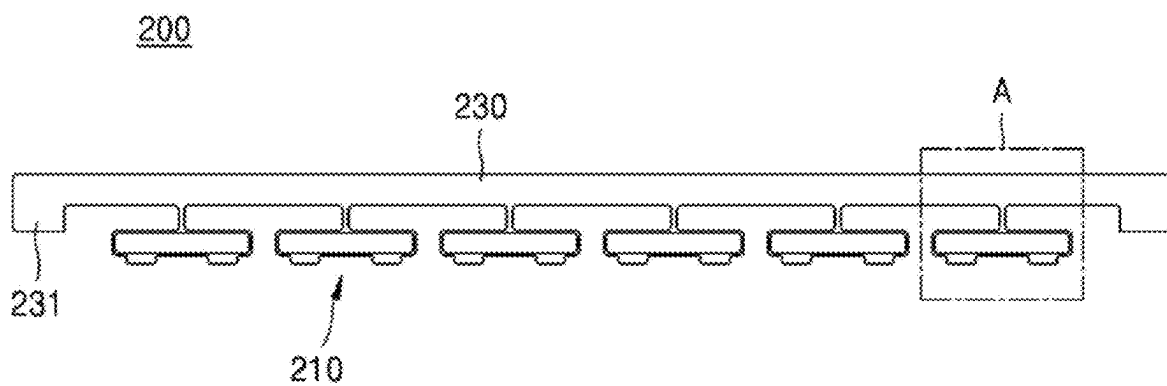


FIG. 5

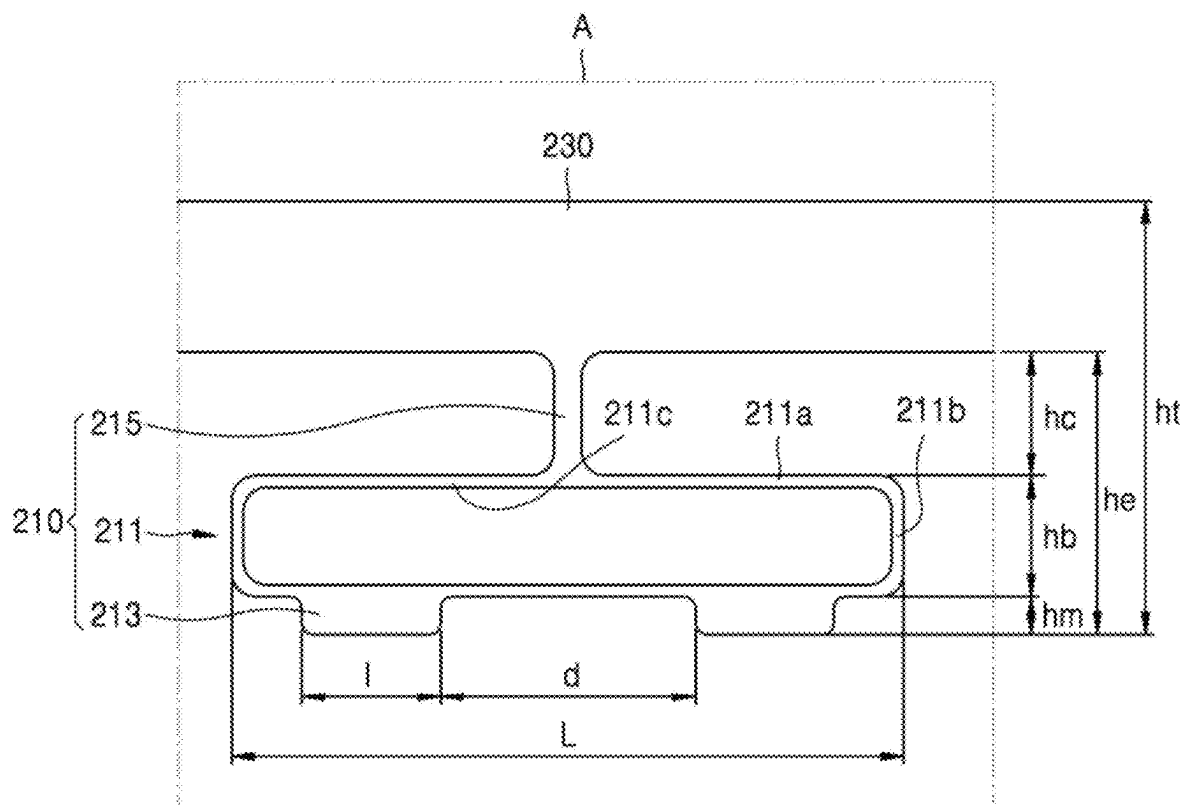


FIG. 6

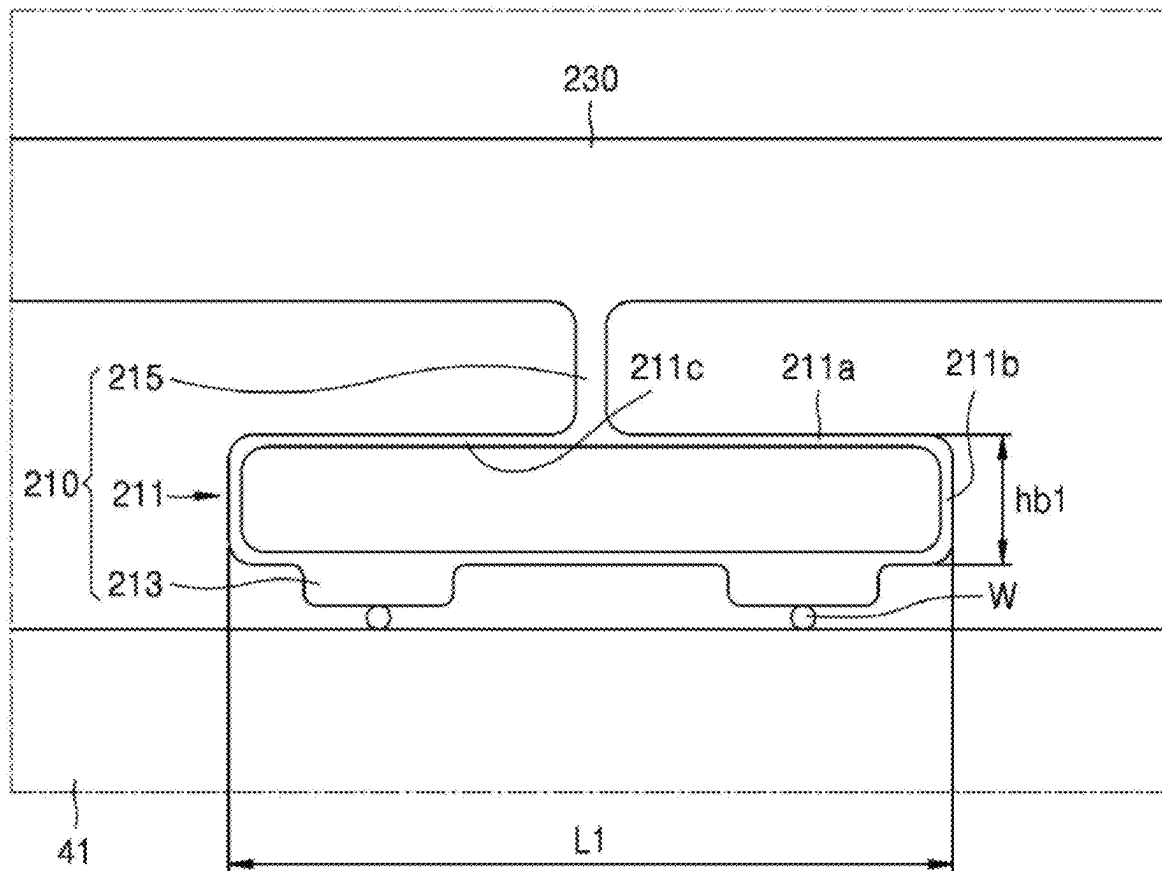


FIG. 7

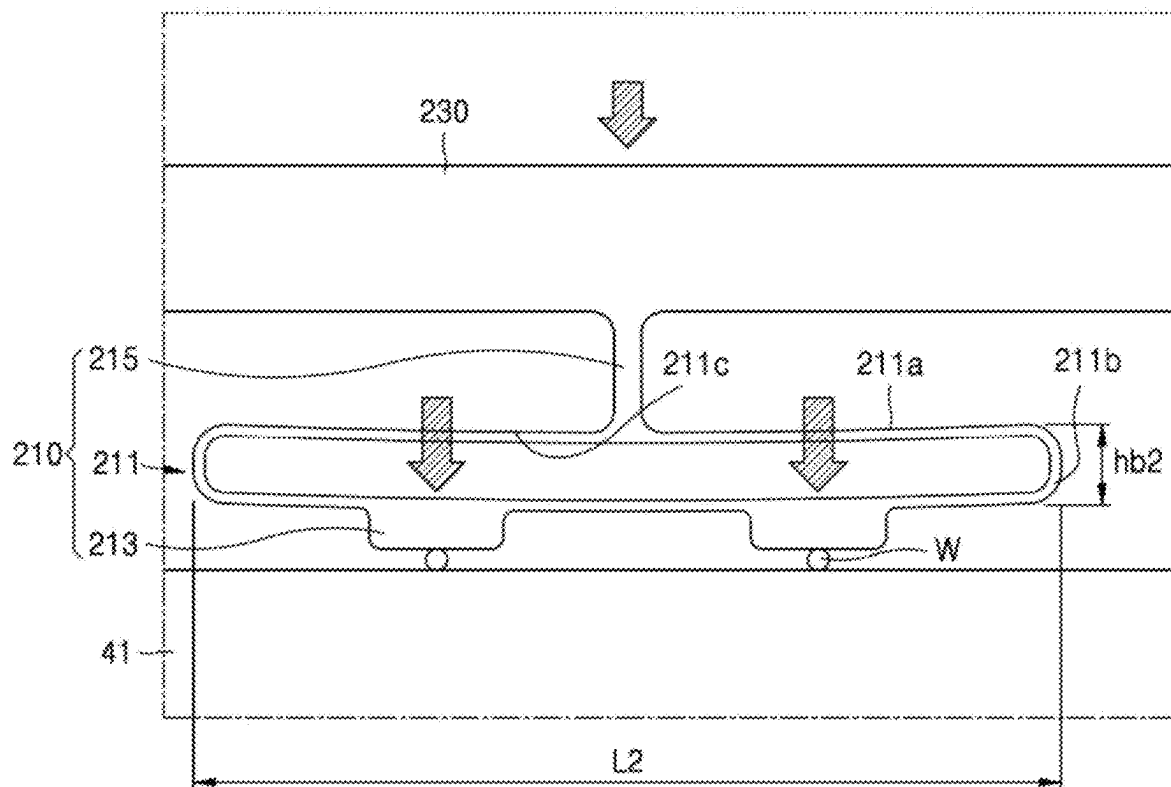


FIG. 8

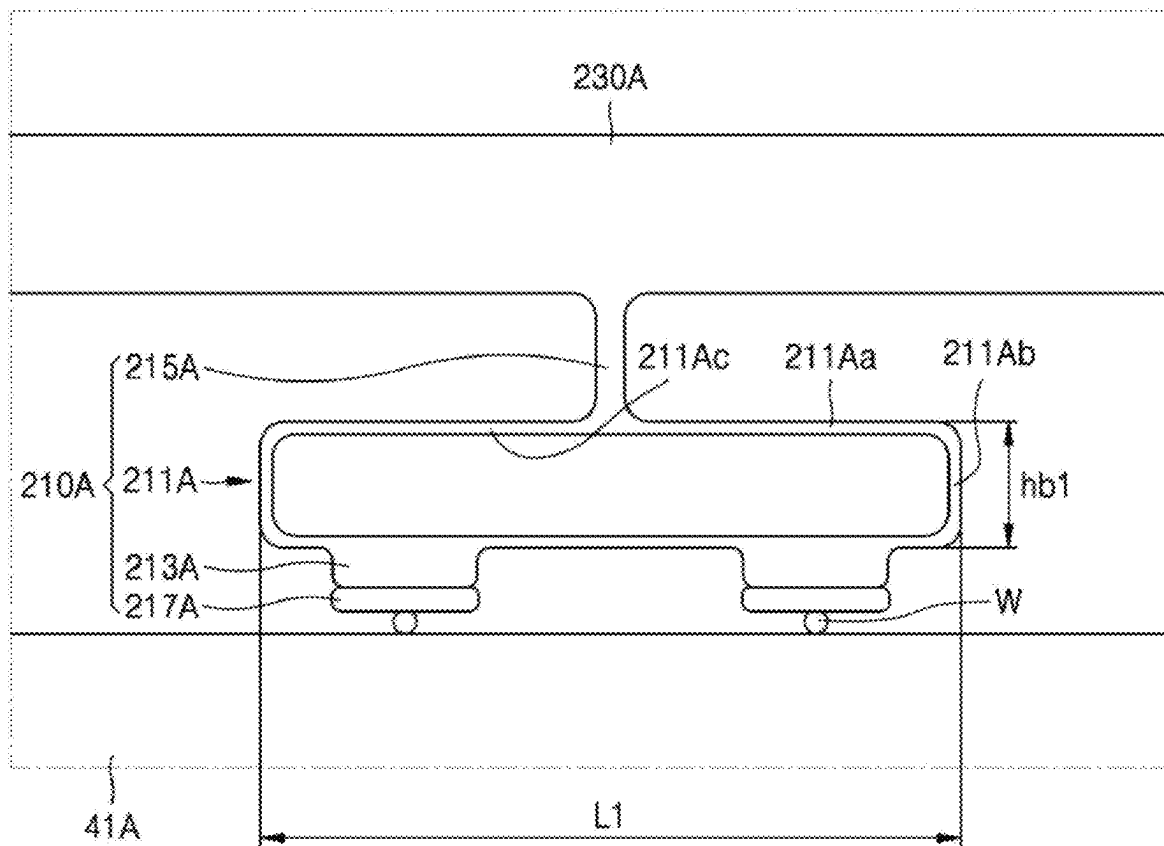
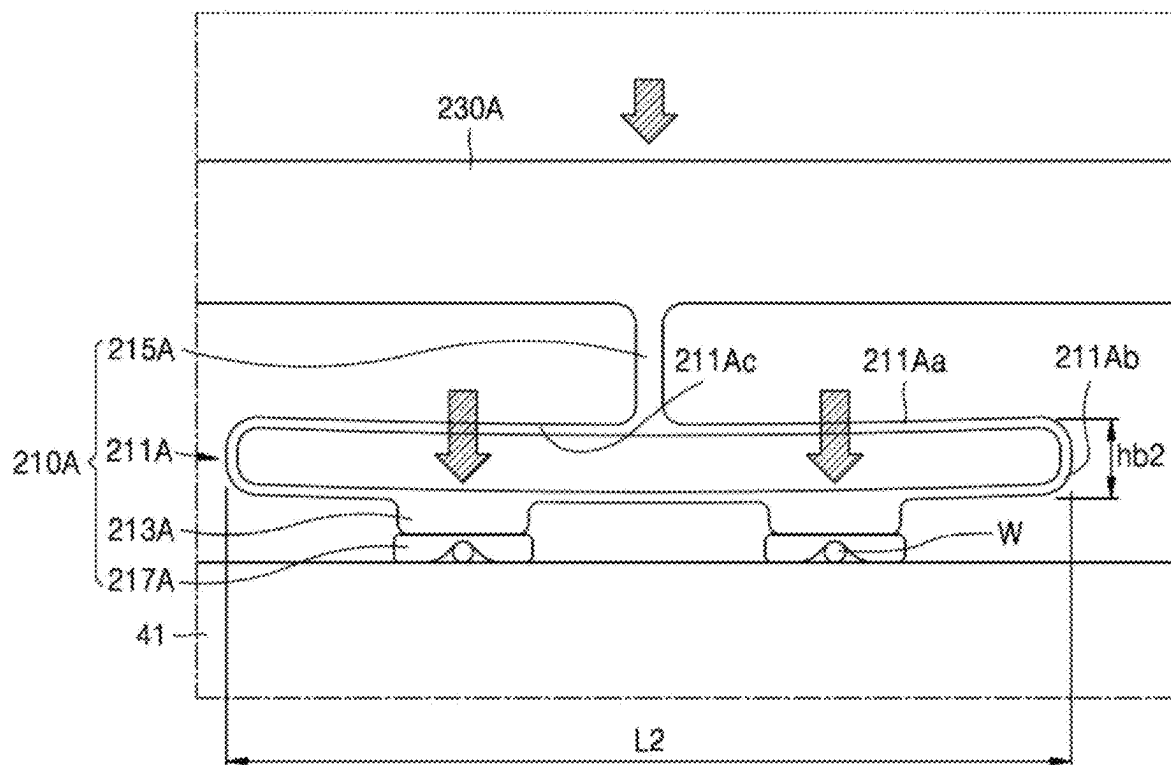


FIG. 9



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WIRE JIG AND TABBING APPARATUS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Korean Patent Application No. 10-2022-0060945, filed on May 18, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

1. Field

Embodiments of the disclosure relates to a wire jig and a tabbing apparatus including the same.

2. Description of the Related Art

A solar cell, or a photovoltaic cell, is formed by arranging a p-n junction diode on a substrate. When a solar light is irradiated onto a solar cell, an exciton that is an electron-hole pair is generated, and as the exciton is separated, an electron is moved to an n layer, and a hole is moved to a p layer, and thus, a photovoltaic force is generated in the p-n junction. Tabbing is a process of arranging a wire on a plurality of solar cells and electrically connecting the solar cells to each other, thereby forming one solar cell module.

A tabbing apparatus generally includes a transfer apparatus for transferring a wire, and bonds a solar cell and a wire in a state in which the solar cell is arranged above the wire. In this process, the wire may be displaced, for example, the wire may be lifted, detached from a designated position, and the like, and thus, the wire and the solar cell may not be connected to each other at a desired position or the connection state therebetween may be poor.

The background technology described above is technical information that the inventor possessed for the derivation of the disclosure or acquired in the derivation process of the disclosure, and it cannot be said that it is known technology disclosed to the general public before the filing of the disclosure.

SUMMARY

Provided are a wire jig for supporting one or more wires during connecting a solar cell with the one or more wires, to stably connect the solar cell with the one or more wires, and a tabbing apparatus including the wire jig. However, such an object is an example, and the objective to be solved by the disclosure is not limited to thereto.

According to an embodiment, a wire jig may include a wire jig provided in a tabbing apparatus to support at least one wire, the wire jig including a main body portion including a window therein; and at least one support portion arranged to correspond to the window, and including at least one support component, wherein the at least one support component is configured to press the at least one wire.

In the wire jig according to an embodiment, the shape of each of the at least one support component may be configured to vary according to a contact state with the at least one wire.

In the wire jig according to an embodiment, the at least one support component may be configured to be compressed when contacting the at least one wire to press the at least one

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wire, and may be an elastic member restored to an original shape when separated from the at least one wire.

In the wire jig according to an embodiment, each of the at least one support component may include a body having an opening therein, and at least one contact member arranged on one side of the body, and configured to contact the at least one wire, and at least one of the body and the at least one contact member is an elastic member.

In the wire jig according to an embodiment, the body may have a ring shape forming a closed contour surrounding the opening.

In the wire jig according to an embodiment, the body may have a pair of long sides in the length direction of the wire jig, and a pair of short sides in the height direction of the wire jig, and the at least one contact member may be arranged on a lower long side of the body.

In the wire jig according to an embodiment, each of the at least one support portion may further include a guide extending in the window, each of the at least one support component may include a connector to connect the guide with the body, and at least one of the guide and the connector may be an elastic member.

In the wire jig according to an embodiment, the at least one contact member may include a plurality of contact members, and the connector may have the length shorter than the body, and may be arranged between the plurality of contact members.

In the wire jig according to an embodiment, the at least one support component may include a plurality of support components arranged spaced apart from each other in the length direction of the wire jig, and the at least one contact member may include a plurality of contact members spaced apart from each other in one of the at least one support component, each of the plurality of contact members contacting the at least one wire.

In the wire jig according to an embodiment, the height of each of the at least one support component may be 50% to 75% of the height of each of the at least one support portion.

In the wire jig according to an embodiment, the height of each of the at least one contact members may be 10% to 40% of the length of one of the pair of long sides of the body.

In the wire jig according to an embodiment, the at least one support portion may include a plurality of support portions arranged spaced apart from each other in the width direction of the wire jig.

In the wire jig according to an embodiment, the at least one wires may include a plurality of wires arranged on a conveyor to be spaced apart from each other, and the plurality of support portions may simultaneously press the plurality of wires to be arranged between an upper surface of the conveyor and the plurality of support portions.

According to an embodiment, a tabbing apparatus for connecting a solar cell with at least one wire includes a wire jig for supporting the at least one wire and a transport apparatus for moving the wire jig, wherein the wire jig may include a main body portion including a window therein, and at least one support portion arranged to correspond to the window and including at least one support component, and the at least one support component is configured to press the at least one wire.

In the wire jig according to an embodiment, the tabbing apparatus may further include a sensor for detecting a contact state between the at least one support portion and the at least one wire, and the transport apparatus may maintain the contact state between the at least one support portion and the at least one wire based on the contact state detected by the sensor.

Other aspects, features, and advantages than those described above will become apparent from the following drawings, claims, and detailed description of the disclosure

BRIEF DESCRIPTION OF DRAWINGS

These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:

FIG. 1 schematically illustrates a tabbing apparatus according to an embodiment;

FIG. 2 is a perspective view of a front side of a wire jig according to an embodiment;

FIG. 3 is a perspective view of a rear side of a wire jig according to an embodiment;

FIG. 4 is a side view of a support portion according to an embodiment;

FIG. 5 is an enlarged side view of a support component according to an embodiment;

FIGS. 6 and 7 illustrate an operation of a wire jig according to an embodiment; and

FIGS. 8 and 9 illustrate an operation of a wire jig according to another embodiment.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. The embodiments are example embodiments, and thus, the embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. As used herein, an expression “at least one of” preceding a list of elements modifies the entire list of the elements and does not modify the individual elements of the list. For example, an expression, “at least one of a, b, and c” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c.

As the disclosure allows for various changes and numerous embodiments, embodiments will be illustrated in the drawings and described in detail in the written description. However, this is not intended to limit the disclosure to particular modes of practice, and it is to be appreciated that all changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the disclosure are encompassed in the disclosure. In the description of the embodiments, even though illustrated in other embodiments, the same identification numbers are used for the same components.

In the following embodiments, while terms such as “first,” “second,” etc., may be used to describe various elements, such elements must not be limited to the above terms, and, the expression of singularity in the specification includes the expression of plurality unless clearly specified otherwise in context.

Sizes of elements in the drawings may be exaggerated for convenience of explanation. For example, since sizes and thicknesses of elements in the drawings are arbitrarily illustrated for convenience of explanation, the following embodiments are not limited thereto.

When a certain embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order.

FIG. 1 schematically illustrates a tabbing apparatus 1 according to an embodiment; FIG. 2 is a perspective view of a front side of a wire jig 10 according to an embodiment; FIG. 3 is a perspective view of a rear side of the wire jig 10 according to an embodiment; FIG. 4 is a side view of a support portion 200 according to an embodiment; FIG. 5 is an enlarged side view of a support component 210 according to an embodiment; and FIGS. 6 and 7 illustrate an operation of the wire jig 10 according to an embodiment.

The tabbing apparatus 1 connects a solar cell C with a wire W to form a solar cell module. The solar cell C includes a semiconductor junction region having a p-n junction surface, and when energy of a certain amount or more is irradiated to the solar cell C, an electromotive force is generated to convert light energy into electrical energy. A semiconductor material included in the solar cell C is not particularly limited, and may include silicon (single crystal, polycrystalline, and amorphous), gallium arsenide, cadmium tellurium, cadmium sulfide, indium phosphorus, copper indium gallium selenide (CIGS), organic dye, or a mixture thereof.

The wire W is a conductor for electrically connecting a plurality of solar cells C to each other, and connects a front surface of a solar cell C to a rear surface of an adjacent solar cell C. For example, the wire W may be electrically connected to the solar cell C through a soldering process. Alternatively or additionally, the wire W may be electrically connected to the solar cell C by using an electrical conductive adhesive (ECA).

In an embodiment, the diameter of the wire W may be 0.1 mm to 0.5 mm. For example, the diameter of the wire W may be 0.15 mm to 0.45 mm. As another example, the diameter of the wire W may be 0.2 mm to 0.4 mm. As yet another example, the diameter of the wire W may be 0.26 mm to 0.38 mm.

In an embodiment, as illustrated in FIG. 1, the tabbing apparatus 1 may include a wire jig 10, a solar cell transfer apparatus 20, a solar cell separation apparatus 30, a wire transfer apparatus 40, a transport apparatus 50, a bonding apparatus 60, and a wire jig transfer apparatus 70.

The wire jig 10 is input to the tabbing apparatus 1 by the wire jig transfer apparatus 70 arranged on one side of the tabbing apparatus 1. The wire jig 10 holds the position of the wire W during a tabbing process so that the solar cell C may be well coupled to the wire W. The wire jig 10 is described herebelow.

The solar cell transfer apparatus 20 externally receives and transfers the solar cell C. For example, the solar cell transfer apparatus 20 including a conveyor may move the solar cell C to the solar cell separation apparatus 30. The solar cell C is in a state of not being separated (scribed) into a certain size.

The solar cell separation apparatus 30 may separate the solar cell C received from the solar cell transfer apparatus 20 in a necessary size. For example, the solar cell separation apparatus 30 includes a scribe 31, and the scribe 31 may separate the solar cell C into a plurality of segments by using laser or a mechanical method.

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The wire transfer apparatus **40** externally receives and moves one or more wires **W**. For example, the wire transfer apparatus **40** may include a conveyor **41**, a roller **43**, and an aligner **45**.

The conveyor **41** may have a flat upper surface so that one or more wires **W** may move in a state of being spaced apart from each other. The roller **43** is arranged in an end portion of the wire transfer apparatus **40**, and transfers the wire **W** in a state of being placed on the upper surface of the conveyor **41**. In an embodiment, the roller **43** may be arranged on each of one side and the other side of the wire transfer apparatus **40** such that the conveyor **41** is arranged therebetween. The aligner **45** primarily aligns the wire **W** before the solar cell **C** contacts the wire **W**, that is, the solar cell **C** is arranged on the wire **W**. For example, the aligner **45** has grooves or protrusions as many as the number corresponding to the wires **W** so that the wire **W** may move along a designated path.

The transport apparatus **50** moves the solar cell **C** and the wire jig **10** to the wire **W**. In an embodiment, the transport apparatus **50** may include a first frame **51**, a pair of second frames **53**, and a transport head **55**. The first frame **51** and the second frames **53** are arranged to cross each other, and the first frame **51** may be movably arranged between the second frames **53**. Furthermore, the transport head **55** is arranged on one side of the first frame **51** to be movable along the first frame **51**, and may pick up an object by an absorption method, a method using an electromagnetic force, or a mechanical method.

In an embodiment, while holding the solar cell **C** and the wire jig **10**, the transport head **55** moves to the wire **W**, and may place the solar cell **C** and the wire jig **10** thereon. For example, as illustrated in FIG. 1, the transport head **55** picks up the solar cell **C** separated by the solar cell separation apparatus **30**, and may arranged the solar cell **C** above the wire **W**. Furthermore, the transport head **55** picks up the wire jig **10** transferred by the wire jig transfer apparatus **70**, and may place the wire jig **10** above the wire **W**. For example, the transport head **55** makes the wire jig **10** contact the wire **W** to maintain the wire **W** at a fixed position, and then, positions the solar cell **C** on the wire **W**.

In an embodiment, while maintaining the wire **W**, the wire jig **10** may move a certain distance or more with the wire **W** in the wire transfer apparatus **40**.

The bonding apparatus **60** may bond the solar cell **C** and the wire **W** contacting each other. For example, the bonding apparatus **60** may be a soldering apparatus that coats and heats solder in a connection portion between the solar cell **C** and the wire **W**, and bonds the same. Alternatively or additionally, the bonding apparatus **60** may be an apparatus that coats an electrically conductive adhesive in a connection portion between the solar cell **C** and the wire **W** and bonds the same.

Referring back to FIGS. 1 to 7, the wire jig **10** according to an embodiment is described.

The wire jig **10** may include a main body portion **100** and a support portion **200**.

The main body portion **100** forms an overall frame of the wire jig **10**, and the support portion **200** may be arranged on one side of the main body portion **100**. The shape of the main body portion **100** is not particularly limited, and for example, as illustrated in FIG. 2, may have a largely thin rectangular prism shape.

In an embodiment, the main body portion **100** may include a window **110** inside. The window **110** may be arranged to overlap at least part of an area in which a plurality of support portions **200** described below are

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arranged. Accordingly, even when the support portion **200** presses the wire **W** so that the shape of the support portion **200** is deformed to be displaced in the height direction, the support portion **200** may not interfere with the main body portion **100**. In an embodiment, the window **110** may be formed between opposite ends of the main body portion **100** in the length direction.

The main body portion **100** may include a support rib **130**. As illustrated in FIG. 2, the support rib **130** may be arranged between the opposite ends of the main body portion **100** in the length direction, and may define the window **110** therein. In an embodiment, the support rib **130** may be provided in a pair to be spaced apart from each other in the width direction. The support rib **130** may prevent the wire jig **10** from being twisted or damaged as the wire jig **10** presses the wire **W**. In an embodiment, each support rib **130** may be arranged not to overlap the support portion **200**. In other words, a plurality of support portions **200** may be arranged between the support ribs **130**.

The main body portion **100** may include an insertion groove **150**. As illustrated in FIG. 3, the insertion groove **150** may be formed inside the main body portion **100**, for example, in inner sides of the opposite ends of the main body portion **100** in the length direction. An insertion protrusion **231** of the support portion **200** described below may be inserted in the insertion groove **150**. For example, the insertion groove **150** is open at the bottom and closed at the top. Accordingly, the support portion **200** may be inserted into the insertion groove **150** under the main body portion **100**.

In an embodiment, the insertion groove **150** may have a shape with a width decreasing downwardly in FIG. 2 (upwardly in FIG. 3). In other words, the insertion groove **150** may have a V shape downwardly in the thickness direction of the main body portion **100**. Accordingly, after the support portion **200** is inserted into the insertion groove **150**, the support portion **200** may not be detached in a process in which the wire jig **10** presses the wire **W**.

The insertion groove **150** may be provided in the same number as the support portion **200**, and a pair of insertion grooves **150** may be arranged to face each other with respect to the window **110**.

The support portion **200** supports the wire **W** according to an operation of the wire jig **10**. For example, the wire jig **10** is picked up by the transport head **55** of the transport apparatus **50**, moved to a designated position on the wire **W**, and descends to press and support the wire **W**. The support portion **200** prevents the wire **W** from being lifted or deviated from the designated position in the transfer process, thereby facilitating the bonding between the solar cell **C** and the wire **W**.

In an embodiment, the support portion **200** may be arranged to correspond to one side of the main body portion **100**, for example, the window **110**. For example, as illustrated in FIGS. 2 and 3, the support portion **200** may include one or more support portions, and may be arranged in an area corresponding to the window **110**.

In an embodiment, the support portion **200** may include a plurality of support portions. For example, as illustrated in FIGS. 2 and 3, the support portion **200** may include a plurality of support portions arranged spaced apart from each other in the width direction of the wire jig **10**. Accordingly, as the support portions **200** are arranged spaced apart a certain distance from each other in the direction of the wire **W**, the support portions **200** may support the wire **W** in a large area.

In an embodiment, the height of the support portion **200** may be 10 mm to 30 mm. For example, the height of the support portion **200** may be 15 mm to 25 mm. As another example, the height of the support portion **200** may be 17 mm to 20 mm. As yet another example, the height of the support portion **200** may be 18.62 mm.

In an embodiment, the length of the support portion **200** may be 100 mm to 300 mm. For example, the length of the support portion **200** may be 150 mm to 250 mm. As another example, the length of the support portion **200** may be 190 mm to 200 mm. As yet another example, the length of the support portion **200** may be 195.75 mm.

In an embodiment, the thickness of the support portion **200** may be 0.1 mm to 3 mm. For example, the thickness of the support portion **200** may be 0.5 mm to 2 mm. As another example, the thickness of the support portion **200** may be 0.7 mm to 1.5 mm. As yet another example, the thickness of the support portion **200** may be 1 mm.

In an embodiment, the shape of the support portion **200** may be deformed by contacting the wire W. For example, the support portion **200** may be an elastic member.

In an embodiment, the material of the support portion **200** may be a metal. For example, the support portion **200** may include alloy steel such as high speed steel that is a kind of high-speed tool steel, not being limited thereto. For example, the support portion **200** may include SKH9 high speed steel. In an embodiment, a support component **210** and a guide **230** of the support portion **200** described below may include the same material.

In an embodiment, the elastic coefficient of the support portion **200** may be 10*10⁶ psi to 100*10⁶ psi. For example, the elastic coefficient of the support portion **200** may be 20*10⁶ psi to 60*10⁶ psi. As another example, the elastic coefficient of the support portion **200** may be 30*10⁶ psi to 50*10⁶ psi. As yet another example, the elastic coefficient of the support portion **200** may be 40*10⁶ psi.

In an embodiment, the support portion **200** may include the support component **210** and the guide **230**.

The support component **210** directly supports the wire W. For example, the support component **210** includes one or more support components arranged in the support portion **200**, and as the wire jig **10** descends toward the wire W, the support component **210** contacts and presses the wire W against the upper surface of the conveyor **41**.

In an embodiment, the support component **210** may include a plurality of support components. As illustrated in FIGS. **2** and **3**, the support component **210** may include a plurality of support components arranged spaced apart from each other in the length direction of the wire jig **10**. Each support component **210** may support a different wire W. Although six support components are illustrated as the support component **210** in the drawings, the number of support components is not particularly limited, and for example, the support component **210** may include two or more support components.

In an embodiment, the support component **210** may support one or more wires W. For example, one support component **210** may be arranged to correspond to two neighboring wires W.

In an embodiment, the support component **210** may have a shape that varies according to a contact state with the wire W. For example, when not contacting the wire W, the support component **210** may maintain the original shape. When contacting the wire W, the shape of the support component **210** may be deformed according to a contact force. Accordingly, the support component **210** may press the wire W harder with a force to restore the original shape.

In an embodiment, the support component **210** may be an elastic member. For example, the support component **210** is compressed when contacting the wire W and may press the wire W with a restoring force thereof. Furthermore, when separated from the wire W, the support component **210** may be restored to the original shape.

In an embodiment, a height h_e of the support component **210** may be 50% or more of a height h_t of the support portion **200**. For example, the height h_e of the support component **210** may be 55% to 75% of the height h_t of the support portion **200**. When the height h_e of the support component **210** is less than 50% of the height h_t of the support portion **200**, it may be difficult to obtain effective elastic deformation when the support component **210** presses the wire W. Furthermore, when the height h_e of the support component **210** exceeds 75% of the height h_t of the support portion **200**, the height h_e of the support component **210** is too high, and thus, the reliability of a wire supporting process may be deteriorated, for example, the support component **210** is excessively bent when pressing the wire W and the like. As another example, in order to secure the reliability of the support component **210** and effectively fix the wire W, the height h_e of the support component **210** may be 60% to 70% of the height h_t of the support portion **200**.

In an embodiment, the support component **210** may include a body **211** and a contact member **213**.

The body **211** has an opening **211c** therein, and may be deformed by contacting the wire W. For example, when contacting the wire W, the body **211**, as an elastic member, may be deformed in the length, height, and width directions. In an embodiment, the body **211** may have a closed contour surrounding the opening **211c**.

The shape of the body **211** is not particularly limited, and in an embodiment, the body **211** may include a pair of long sides **211a** and a pair of short sides **211b**. For example, as illustrated in FIG. **5**, the body **211** may have a rectangular shape having a pair of the long sides **211a** in the length direction of the wire jig **10**, and a pair of the short sides **211b** in the height direction of the wire jig **10**. Before the deformation of the body **211**, the length of each of the long sides **211a** may be L, and the length of each of the short sides **211b** may be h.

In an embodiment, in order to secure reliability according to the elastic deformation of the body **211** when pressing the wire W, a height h_b of the body **211** may be 50% or less of the height h_e of the support component **210**.

The contact member **213** is arranged on one side of the body **211** to directly contact the wire W. For example, the contact member **213** may include one or more contact members arranged on a lower side of the body **211**, for example, in a lower one of a pair of the long sides **211a** of the body **211**, to protrude toward the wire W.

In an embodiment, the contact member **213** may include a plurality of contact members. For example, as illustrated in FIG. **5**, the contact member **213** may include two contact members arranged on a lower end portion of the body **211** to be spaced apart from each other by a distance d in the length direction of the wire jig **10**. Furthermore, the contact members **213** may each correspond to one wire W. In other words, when the wires W are arranged on the conveyor **41**, the contact members **213** included in one support component **210** may be in contact with different wires W.

In an embodiment, the contact member **213** may have a length l. The length l of the contact member **213** may be shorter than the length of each of the long sides **211a** of the body **211**, and furthermore, shorter than the distance d between the contact members **213** neighboring each other.

For example, the length l of the contact member **213** may be 5 times to 50 times of the diameter of the wire W . For example, the length l of the contact member **213** may be 10 times to 30 times of the diameter of the wire W . When the length l of the contact member **213** is less than 10 times of the diameter of the wire W , during the lowering of the wire jig **10**, it is difficult to arrange the wire W in a central area of the contact member **213** so that it may be difficult to effectively support the wire W . When the length l of the contact member **213** exceeds 30 times of the diameter of the wire W , the weight of the support component **210** increases so that a load applied to a connector **215** increases. Accordingly, as the process is repeated, the support portion **200**, in particular, the connector **215**, may be damaged.

In an embodiment, the length l of the contact member **213** may be 10% to 40% of the length L of each of the long sides **211a** of the body **211**. For example, the length l of the contact member **213** may be 15% to 35% of the length L of each of the long sides **211a** of the body **211**. When the length l of the contact member **213** is less than 10% of the length L of each of the long sides **211a**, the length l of the contact member **213** is decreased too short so that it may be difficult to effectively press the wire W . When the length l of the contact member **213** exceeds 40% of the length L of each of the long sides **211a**, the weight of the support component **210** increases so that the load applied to the connector **215** increases. Accordingly, as the process is repeated, the support portion **200**, in particular, the connector **215**, may be damaged.

In an embodiment, the length l of the contact member **213** may be 75% to 95% of the distance d between the contact members **213**. When the length l of the contact member **213** is less than 75% of the distance d between the contact members **213**, it may be difficult for the contact member **213** to effectively press the wire W . When the length l of the contact member **213** exceeds 95% of the distance d between the contact members **213**, the weight of the support component **210** increases so that the load applied to the connector **215** increases. Accordingly, as the process is repeated, the support portion **200**, in particular, the connector **215**, may be damaged.

In an embodiment, the height h_m of the contact member **213** may be 20% or less of the height h_e of the support component **210**. In an embodiment, the height h_m of the contact member **213** may be less than the height h_b of the body **211**. Accordingly, the reliability of the support component **210** may be secured, that is, when pressing the wire W , the support component **210** is not deviated from the designated position and is not excessively deformed, and may be firmly support the wire W .

In an embodiment, the body **211** and/or the contact member **213** may be an elastic member. The body **211** and/or the contact member **213** includes a material that is deformable and elastic. Accordingly, when the wire jig **10** contacts the wire W , the body **211** and/or the contact member **213** are deformed so as to press the wire W harder.

In an embodiment, the support component **210** may further include the connector **215**. As illustrated in FIG. 5, in order to connect the support component **210** with the guide **230**, one side of the connector **215** may be connected to the body **211**, and the other side thereof may be connected to the guide **230**.

In an embodiment, the connector **215** may be arranged between the contact members **213**. For example, as illustrated in FIG. 5, while the two contact members **213** are arranged on one of the long sides **211a** of the body **211**, one side of the connector **215** may be arranged between the two

contact members **213**. For example, the connector **215** may be arranged in the middle of the two contact members **213** at the other of the long sides **211a**. Accordingly, the support component **210** may uniformly press the wires W .

In an embodiment, the connector **215** may be an elastic member. In other words, when the wire jig **10** contacts the wire W , in addition to the deformation of the body **211** and/or the contact member **213**, the connector **215** may be compressed in the length direction to press the wire W harder.

In an embodiment, to secure reliability according to the elastic deformation of the connector **215** when pressing the wire W , the height h_c of the connector **215** may be greater than 50% or less of the height h_e of the support component **210**, and the height h_c of the connector **215** may be greater than the height h_b of the body **211**. For example, the height h_b of the body **211** may be 90% or more of the height h_c of the connector **215**. For example, the height h_b of the body **211** may be 95% or more of the height h_c of the connector **215**.

In an embodiment, the support component **210** may include a metal material, for example, SKH9 high speed steel. Furthermore, the body **211**, the contact member **213**, and the connector **215** of the support component **210** may all include the same material and may be formed integrally.

The guide **230** is arranged on one side of the main body portion **100** to connect the main body portion **100** with the support component **210**. In an embodiment, as illustrated in FIGS. 2 to 4, the guide **230** has a rod or thin plate shape extending in the length direction of the wire jig **10**, and opposite ends thereof may be connected to inner sides of the opposite ends of the main body portion **100**.

In an embodiment, the guide **230** may include a plurality of guides. For example, as illustrated in FIGS. 2 and 3, the guide **230** may include a plurality of guides arranged spaced apart from each other in the width direction of the wire jig **10**.

In an embodiment, the guide **230** may be detachably connected to the main body portion **100**. For example, as illustrated in FIG. 4, the guide **230** may include the insertion protrusion **231** in the opposite ends in the length direction thereof. As the insertion protrusion **231** protrudes downwardly from the opposite ends of the guide **230**, the guide **230** may come in contact with a large area of the insertion groove **150**. Furthermore, the guide **230** may be detachably inserted into the insertion groove **150**. Accordingly, considering the number, physical properties, and the like of the wires W , the numbers of the guides **230** included in the wire jig **10** and the support components **210** according thereto may be adjusted.

In an embodiment, the guide **230** may be an elastic member. In other words, when the wire jig **10** contacts the wire W , in addition to the deformation of the body **211** and/or the contact member **213**, the guide **230** may be compressed in the length direction so as to press the wire W harder.

Next, the operation of the wire jig **10** according to an embodiment is described with reference to FIGS. 1 to 7.

As illustrated in FIG. 1, the solar cell C is transferred by the solar cell transfer apparatus **20** to the solar cell separation apparatus **30** and separated in an appropriate size. Simultaneously or with a time gap, the wire W is transferred by the wire transfer apparatus **40** on the conveyor **41**. Furthermore, simultaneously or with a time gap, the wire jig **10** is transferred by the wire jig transfer apparatus **70**. The transport head **55** of the transport apparatus **50** is moved to a preset position through the first frame **51** and the second

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frames **53** to pick up the wire jig **10**. Then, the transport head **55** is moved to the conveyor **41** in which the wire **W** is placed and positions the wire jig **10** on the wire **W**.

For example, as illustrated in FIG. 6, the transport head **55** is moved to a preset position so that the wire jig **10** may contact the wire **W**. In an embodiment, the tabbing apparatus **1** may include a sensor **S** to check a contact state with the wire **W**. For example, the sensor **S** may be arranged in the transport head **55** and/or the wire jig **10**. The sensor **S** may detect a contact state between the support portion **200** and the wire **W**.

The sensor **S** may be an electromagnetic sensor to check the contact state through electrical connection with the wire **W** that is a conductor. Alternatively or additionally, the sensor **S** may be a pressure sensor to check the contact state by measuring a pressure between the wire **W** and the support portion **200**. For example, when a current detected by the sensor **S** is a certain value or more, the tabbing apparatus **1** may determine that the contact state of the wire **W** and the wire jig **10** is good. Alternatively, when a pressure detected by the sensor **S** is a certain value or more, the tabbing apparatus **1** may determine that the contact state of the wire **W** and the support portion **200** is good. Furthermore, the transport apparatus **50** may maintain the contact state between the support portion **200** and the wire **W** based on the contact state detected by the sensor **S**.

Although the drawings illustrate that one support component **210** contacts two wires **W**, the disclosure is not limited thereto. When one of the support components **210** contacts the wire **W**, the other support components **210** may contact the wire **W**. In other words, the support components **210** may simultaneously contact the wire **W** corresponding thereto.

Next, the tabbing apparatus **1** moves the transport head **55** downwardly so that the wire jig **10** may press the wire **W**. For example, as illustrated in FIG. 7, when the support component **210** is in contact with the wire **W**, the transport head **55** makes the wire jig **10** press the wire **W** downwardly, that is, in a direction perpendicular to the ground or the conveyor **41**. Accordingly, as the shape of the support component **210** changes, the support component **210** presses the wire **W** on the conveyor **41** to support the wire **W**.

For example, as illustrated in FIG. 7, as the body **211** that is an elastic member receives a force in a vertical direction, the long sides **211a** of the body **211** are bent downwardly so that the length **L** of each of the long sides **211a** of the body **211** is changed to **L2**, and a length **hb1** of each of the short sides **211b** is changed to a length **hb2**. As the shape of the body **211** changes, the wire **W** may be firmly supported by a restoring force of the body **211** to return to the original shape. Furthermore, when the support portion **200** is entirely formed of an elastic member, elements other than the body **211** may be elastically deformed by contacting the wire **W**. For example, the contact member **213**, the connector **215**, and the guide **230** may be elastically deformed in a direction perpendicular to the height direction through the contact with the wire **W**.

Although the drawings illustrate that only the body **211** is an elastic member, the disclosure is not limited thereto. As described above, in addition to the body **211**, the contact member **213**, the connector **215**, and/or the guide **230** may be elastic members. Accordingly, as the shapes of the contact member **213**, the connector **215**, and/or the guide **230** change, the wire **W** may be firmly supported by a restoring force of the body **211**.

In an embodiment, the transport head **55** may uniformly maintain a pressurization state of the wire **W** through the

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sensor **S**. For example, to maintain the current value or pressure value detected by the sensor **S** within a preset range, the transport head **55** may press the wire jig **10** against the wire **W**.

Then, when the process of placing the solar cell **C** on the wire **W** is complete, the transport head **55** picks up the wire jig **10** to remove the pressurization state. Then, the solar cell **C** and the wire **W** are moved by the wire transfer apparatus **40** to the bonding apparatus **60**, and electrically connected to each other forming a solar cell module.

Through the above configuration, the wire jig **10** and a tabbing apparatus **1** including the wire jig **10**, according to an embodiment, prevent the lifting of the wire **W** or the displacement of the wire **W** during the connection of the solar cell **C** and the wire **W**, thereby facilitating the connection of the solar cell **C** and the wire **W** at the designated position.

FIGS. 8 and 9 illustrate an operation of the wire jig **10** according to another embodiment.

The wire jig **10** according to the present embodiment may have a support component **210A**, and a configuration of the support component **210A** may be different from a configuration of the support component **210** according to the embodiment described above. For example, the support component **210A** may further include a reinforcement member **217A**. The other configuration of the support component **210A** may be the same as the configuration of the support component **210**, detailed descriptions thereof are omitted.

As illustrated in FIGS. 8 and 9, the support component **210A** may further include the reinforcement member **217A** in an end portion of a contact member **213A**. The reinforcement member **217A** may be arranged on one surface of the contact member **213A**, for example, a lower surface facing the wire **W**. The reinforcement member **217A** may be arranged to correspond to each contact member **213A**. The reinforcement member **217A** may have the same area as the contact member **213A**. Alternatively, the reinforcement member **217A** may have an area greater than or smaller than the contact member **213A**.

In an embodiment, the reinforcement member **217A** may include a material that is deformable and flexible. For example, the reinforcement member **217A** may include a material, such as rubber, silicon, and the like, being deformable and having a high friction coefficient. Accordingly, when the wire jig **10** presses the wire **W**, the wire **W** may be surely prevented from being deviated from the designated position.

While the reinforcement member **217A** is in contact with the wire **W**, when the wire jig **10** presses the wire **W** downwardly, that is, in the direction perpendicular to the ground or the conveyor **41**, as illustrated in FIG. 9, a body **211A**, the contact member **213A**, a connector **215A**, and/or a guide **230A** are deformed so as to support the wire **W**. In detail, as a long side **211Aa** of the body **211A** is bent downwardly, the length of the long side **211Aa** may be changed from **L1** to **L2**, and the length of a short side **211Ab** may be changed from **hb1** to **hb2**. Furthermore, the contact member **213A**, the connector **215A**, and the guide **230A** may be elastically deformed.

Furthermore, in addition thereto, the reinforcement member **217A** may be deformed to encompass the wire **W**. Accordingly, while the wire jig **10** presses the wire **W**, the wire **W** may be firmly supported, and the wire **W** may be prevented from being deviated from the designated position while being pressed.

In an embodiment, at least a portion of the reinforcement member **217A** in a deformed state may be in contact with the

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upper surface of the conveyor 41. For example, as illustrated in FIG. 9, the opposite ends in the length direction of the reinforcement member 217A are in contact with the upper surface of the conveyor 41, and thus, the reinforcement member 217A may be deformed to encompass the wire W between the upper surface of the conveyor 41 and the reinforcement member 217A. Accordingly, the wire W may be firmly supported.

The wire jig and the tabbing apparatus including the wire jig, according to the above embodiments, may prevent the lifting and positional deviation of the wire when the solar cell is coupled to the wire, thereby facilitating the connection of the solar cell and the wire.

In an actual device corresponding to the above embodiments, a connection between the constituent elements may be indicated by various functional connections, physical connections, or circuit connections that can be replaced or added. Furthermore, no item or component is essential to the practice of the disclosure unless the element is specifically described as “essential” or “critical.”

The use of terms “a” and “an” and “the” and similar referents in the context of describing the disclosure are to be construed to cover both the singular and the plural. Furthermore, recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. Furthermore, the steps of all methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. Embodiments are not necessarily limited by the order of description of the steps. The use of any and all examples, or language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. Furthermore, numerous modifications and adaptations will be readily apparent to those of ordinary skill in this art without departing from the spirit and scope of the disclosure.

What is claimed is:

1. A wire jig provided in a tabbing apparatus to support at least one wire, the wire jig comprising:

a main body comprising a window therein; and
at least one support portion arranged to correspond to the window, and comprising at least one support component,

wherein the at least one support component is configured to press the at least one wire, and

wherein each of the at least one support component comprises:

a body having an opening therein; and

at least one contact member protruding toward the at least one wire from one side of the body, and configured to contact the at least one wire,

wherein at least one of the body and the at least one contact member is an elastic member,

wherein the body has a pair of long sides in a length direction of the wire jig, and a pair of short sides in a height direction of the wire jig,

wherein the at least one contact member is arranged on a lower long side of the body, and

wherein a height of each of the at least one contact member is 10% to 40% of a length of one of the pair of long sides of the body.

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2. The wire jig of claim 1, wherein a shape of each of the at least one support component is configured to vary according to a contact state with the at least one wire.

3. The wire jig of claim 1, wherein the at least one support component is configured to be compressed when contacting the at least one wire to press the at least one wire, and

wherein the at least one support component is an elastic member restored to an original shape when separated from the at least one wire.

4. The wire jig of claim 1, wherein the body has a ring shape forming a closed contour surrounding the opening.

5. The wire jig of claim 1, wherein each of the at least one support portion further comprises a guide extending in the window,

wherein each of the at least one support component comprises a connector to connect the guide with the body, and

wherein at least one of the guide and the connector is an elastic member.

6. The wire jig of claim 5, wherein the at least one contact member comprises a plurality of contact members, and wherein the connector has a length shorter than the body, and is arranged between the plurality of contact members.

7. The wire jig of claim 1, wherein the at least one support component comprises a plurality of support components arranged spaced apart from each other in a length direction of the wire jig,

wherein the at least one contact member comprises a plurality of contact members spaced apart from each other in one of the plurality of support components, and wherein each of the plurality of contact members is configured to contact each of the at least one wire.

8. The wire jig of claim 1, wherein a height of each of the at least one support component is 50% to 75% of a height of each of the at least one support portion.

9. The wire jig of claim 1, wherein the at least one support portion comprises a plurality of support portions arranged spaced apart from each other in a width direction of the wire jig.

10. The wire jig of claim 9, wherein the at least one wire comprises a plurality of wires arranged on a conveyor to be spaced apart from each other, and

wherein the plurality of support portions simultaneously are configured to press the plurality of wires to be arranged between an upper surface of the conveyor and the plurality of support portions.

11. The wire jig of claim 1, wherein the at least one support component is configured to elastically press the at least one wire.

12. A tabbing apparatus for connecting a solar cell with at least one wire, the tabbing apparatus comprising a wire jig for supporting the at least one wire and a transport apparatus for moving the wire jig,

wherein the wire jig comprises:

a main body portion comprising a window therein; and
at least one support portion arranged to correspond to the window, and comprising at least one support component, and

wherein the at least one support component is configured to press the at least one wire, and

wherein the tabbing apparatus further comprises a sensor for detecting a contact state between the at least one support portion and the at least one wire, and

wherein the transport apparatus is configured to maintain the contact state between the at least one support

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portion and the at least one wire, based on the contact state detected by the sensor.

13. The tabbing apparatus of claim **12**, wherein the at least one support component is configured to elastically press the at least one wire.

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